Wladimiro Leone

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Profile

Postdoc at the University of Cagliari working on high-energy astrophysics, gamma-ray bursts, timing analysis, transient-source localization, and quantum-gravity phenomenology. Experienced in scientific software development, spectral and temporal analysis, Monte Carlo methods, and mission-level data processing for space instrumentation.

Personal Information

Date of birth:	6 April 1998
Nationality:	Italian
Languages:	Italian (native), English (C1), Spanish (B1), French (A2)

Education

PhD title *2026*
University of Trento - Curriculum: Space Science and Technology Grade: Excellent
Thesis: *Temporal Analysis of High-Energy Transient Sources - An Adaptive Binning Approach in Time-Domain Astronomy*

Master's Degree in Physics *2022*
University of Trieste - Curriculum: Astrophysics and Cosmology
Thesis: *Contribution of distributed architecture to multi-messenger astrophysics: HERMES mission optimization and Space Weather monitoring thanks to CUBE mission*

Bachelor's Degree in Physics *2019*
University of Trieste
Thesis: *Analysis of GRB 180914B, 190501A and 190530A observed by AGILE and Fermi satellites*

Work experiences

Postdoc position *14/05/2026-14/05/2027*
University of Cagliari
Project: *Development of open-source frameworks for high-energy timing analysis and automatic classification of Gamma-Ray Bursts using Machine Learning techniques within the SMART-TIME project.*

Research Interests

Wladimiro Leone is a postdoctoral researcher in high-energy astrophysics working on transient phenomena, with particular emphasis on Gamma-Ray Bursts, compact objects, and time-domain astrophysics. His research focuses on the development of innovative timing techniques, especially optimized for the low-statistics regime, including adaptive binning methods and time-of-arrival–based cross-correlation techniques. These approaches enable robust measurements of temporal variability, spectral lags, and characteristic variability timescales even in photon-limited data. He has applied these techniques across different areas of high-energy astrophysics, focusing on the study of GRBs. In particular, he conducted studies in multi-messenger astrophysics aimed at quantifying the probability of joint detections between gravitational waves and electromagnetic counterparts within the Einstein Telescope framework. He also exploited GRB spectral lags to test Lorentz Invariance Violation predictions in LIV scenarios such as Quantum Gravity phenomenology. His work further includes pulsar timing studies and the investigation of accreting compact objects operating in both sub- and super-Eddington regimes. These timing techniques also proved pivotal for distributed space-based observatories and nanosatellite missions. In particular, Wladimiro led the first detection of a 34 millisecond pulsar using the ultra-compact, wide-field, non-collimated instrument, the HERMES payload aboard the SpIRIT nanosatellite.

Research Periods Abroad

Masaryk University, Brno, Czech Republic *Jan 2024 – Apr 2024*
Activity: HERMES-SP timing software development

University of Melbourne, Melbourne, Australia *Apr 2024 – Aug 2024*
Activity: Represented the *HERMES-SP* collaboration at the *SpIRIT* Mission Control Centre

University of Zaragoza, Zaragoza, Spain *3–26 Jun 2026*

Activity: Research on Quantum Gravity phenomenology and Lorentz Invariance Violation tests with high-energy astrophysical sources within COST Action CA23130 BridgeQG.

IRAP, Toulouse, France *22 Nov – 5 Dec 2026*
Activity: Development of GRB localization strategies for multi-messenger associations and rapid electromagnetic follow-up.

Scientific Responsibilities and Contributions to International Space Missions

HERMES-SP Mission *2024 – present*
ASI – HORIZON 2020 Project
Role: HERMES-SP SOC
Activities: Lead high-level scientific software developer; development of timing software and localization pipeline; contribution to detector-response calibration and on-orbit validation.

SpIRIT Mission *May 2024 – Aug 2024*
University of Melbourne
Role: Data Processing – Extraction and Analysis
Deliverable: FM1/SpIRIT Status Report submitted to ASI on 10 July 2024

THESEUS mission (planned launch: 2037) *January 2026 – Present*
Contribution to the THESEUS Yellow Book:*Quantum Gravity phenomenology with THESEUS*

NewAthena mission (planned launch: 2037) *April 2026 – Present*
Working Group 4: Compact objects
Working Group 5: Transients and multi-messenger astrophysics

Einstein Telescope (planned start of operations: 2037) *May 2026 – Present*
Division 4 (Multi-messenger astrophysics): Ongoing project - *Quantifying the detection probability of electromagnetic counterparts to gravitational-wave events as a function of the considered localization and monitoring facilities*

Teaching Experience

Physics Laboratory I Assistant *2023*
Bachelor's Degree Programme, Department of Physics, University of Palermo
Academy of Distinction – Astrophysics *2024 – present*
Department of Physics, University of Palermo

Science Communication and Outreach

Progetto Matematica e Realtà *2024 – present*
Department of Mathematics and Computer Science, University of Perugia

Science communication and public outreach through social media *2025–present*
Creation of accessible astrophysics content aimed at communicating scientific concepts and current research to a broad audience.

Awards

Best Poster Award *21 June 2025*
3rd Edition of the National PhD Days in DIN – SST, Palermo, Italy

COST Action CA23130 (BridgeQG) Funding Award *3–26 June 2026*
Bridging High and Low Energies in Search of Quantum Gravity, Zaragoza, Spain

TNA 3rd Calls Funding Award *22 November–05 December 2026*
GRB Localizations for multi-messenger sssociations and rapid follow-up, Toulouse, France

Talks and Seminar Invited Contributions

- **2023** – IASF Journal Club, Palermo, *Contribution of the distributed architecture to multi-messenger astrophysics: HERMES mission optimisation and space weather monitoring via the CUBE mission*
- **2023** – IASF Journal Club, Palermo, *High redshift GRB in the JWST era*
- **2024** – Cosmotea, Brno, Czech Republic, *Temporal analysis of light-curves from transient sources: detectors TOAs lists auto-calibration and evaluation of experimental delays*
- **2024** – QUAGRAP Meeting, Scuola Superiore Meridionale, Napoli, *Temporal analysis of fast variability in transient source light curves: cross-correlation techniques for Quantum Gravity phenomenology with GRBs*
- **2024** – Compact Objects National Congress XIII, Alghero, *The HERMES mission: SpIRIT update*

- **2026** – *Gamma-Ray Bursts: The Great Gig in the Sky, Sexten*, *When Milliseconds Matter: Nanosatellite Timing for GRB Localization and Quantum-Gravity Constraints*
- **2026** – *THESEUS CONFERENCE, Malaga*, *Quantum Gravity with THESEUS: A redshift hunter to probe Lorentz Invariance Violation with Gamma-Ray Bursts*
- **2026** – *GEMS WORKSHOP, Rome*, *Fast X-ray Timing with CubeSats: An Adaptive Approach to the Low-Count Regime*
- **2026** – *INTEGRAL Legacy, Paris*, *Probing Quantum Gravity with Energy-Resolved Timing of Gamma-Ray Bursts and Pulsars*

Conference Talks and Posters Contributions

- **2024** – YSC on Astronomy and Space Physics, online participation, *GRBs localization with nano-satellites constellations*
- **2025** – Celebrating 20 Years of Swift Discoveries, Florence, Italy
- **2025** – 3rd Edition of the National PhD Days in DIN – SST, Palermo, Italy, *GRBs as Beacons for Multi-Messenger Astronomy and LIV Investigation*
- **2026** – The many tones of accretion, to the memory of Tomaso Belloni – Cefalù, Italy, *Kilosecond hard lag in NGC 7456 ULX-1: Tracing propagation and reprocessing through the super-Eddington wind*

Workshop and Scientific Schools

Machine and Deep Learning @ INAF-IASF Milano *15–19 Jan 2024*

Observing Proposals

2025 – *Principal Investigator*, CRAB observation for energy and timing calibration with SpIRIT/HERMES

Publications and Metrics

SAO/NASA Astrophysics Data System (as of September 2026)

- Publications: 15
- Refereed international ISI journals: 11
- Non-refereed publications: 4
- Total citations: 32
- H-index: 3

Research Gate (as of September 2026)

- Publications: 15
- Refereed international ISI journals: 11
- Non-refereed publications: 4
- Total citations: 62
- H-index: 5

Research Expertise

High-energy timing and spectral analysis, Monte Carlo simulations, orbital propagation, event-level data processing, scientific software development, FITS telemetry parsing, time-lag estimation.

Technical Skills

Programming:	Python (expert), Fortran (expert), Bash (advanced), C++ (advanced)
Data analysis tools:	XSPEC, SPEX, ROOT, SGP4, efold, efsearch, h5py, HEASoft, FFT, CCF
Mission software:	Find_orb, gpredict, orbital propagators, timing pipelines, SGP4

Publications

Refereed international journals

- **Leone W.** et al., 2025, *A&A*, 701, A50 — Time domain astrophysics with transient sources: Delay estimate via Cross Correlation Function techniques.
- **Leone W.** et al., 2026, *ApJ*, 1001, 5 — CubeSats Reach the Microsecond X-Ray Domain: Crab Pulsar Timing with SpIRIT/HERMES.
- **Leone W.** et al., 2026, *A&A* (In Press) — Detection of kilosecond hard lags in the new candidate pulsating ULXs in NGC 7456 ULX-1, pending DOI: 10.1051/0004-6361/202661582
- R. Iaria, T. Di Salvo, A. Anitra, C. Miceli, F. Barra, **W. Leone**, et al., 2024, *A&A*, 683, A79 — Confirmation of the presence of a CRSF in the NICER spectrum of X 1822−371.
- C. Maraventano, G. Ghirlanda, L. Nava, T. Di Salvo, **W. Leone**, et al., 2025, *A&A*, 697, 161 — High-energy time lags of Gamma-Ray Bursts.
- A. Anitra, C. Miceli, T. Di Salvo, R. Iaria, N. Degenaar, J. M. Miller, F. Barra, **W. Leone**, et al., 2024, *A&A*, 690, A148 — X-ray view of emission lines in optical spectra: Spectral analysis of the two low-mass X-ray binary systems Swift J1357.2−0933 and MAXI J1305−704.
- G. Ghirlanda, L. Nava, O. Salafia, F. Fiore, R. Campana, R. Salvaterra, A. Sanna, **W. Leone**, et al., 2024, *A&A*, 689, A175 — HERMES: Gamma-ray burst and gravitational-wave counterpart hunter.
- R. Iaria, T. Di Salvo, A. Anitra, C. Miceli, **W. Leone**, et al., 2024, *A&A*, 686, A216 — The puzzling orbital residuals of XTE J1710−281: Is a Jovian planet orbiting the binary system?
- C. Miceli, D. Mata Sánchez, A. Anitra, T. Muñoz-Darias, T. Di Salvo, R. Iaria, A. Marino, **W. Leone**, et al., 2024, *A&A*, 684, A67 — Soft-state optical spectroscopy of the black hole MAXI J1305−704.
- R. Iaria, T. Di Salvo, A. Anitra, F. Barra, A. Sanna, C. Maraventano, C. Miceli, **W. Leone**, et al., 2026, *APJ*, 998, 58 — A mysterious feature in the NICER spectrum of 4U 1820−30: A gravitationally redshifted absorption line?
- A. Anitra, A. Gnarini, T. Di Salvo, R. Iaria, A. Marino, F. Barra, L. Burderi, A. Sanna, L. Marra, S. Bianchi, G. Matt, F. Ursini, C. Miceli, A. Miraval Zanon, **W. Leone**, et al., 2026, A&A, 714, A62 - The first IXPE view of the eclipsing ADC source 4U 1822–37

  R. Iaria, A. Anitra, T. Di Salvo, A. Sanna, F. Barra, L. Burderi, C. Maraventano, S. Caserta, **W. Leone**, C. Miceli, 2026, A&A, accepted for pubblication - 4U 1624-490 as a possible intermediate-mass X-ray binary: constraints from XRISM and NuSTAR spectroscopy, pending DOI:10.1051/0004-6361/202660603

Conference proceedings

- F. Fiore et al., 2025, *arXiv:2502.17952* — HERMES Pathfinder & SpIRIT: a progress report.
- G. Baroni et al., 2024, *SPIE Conf. Ser.*, 13093 — The commissioning and early operations of the high-energy HERMES payload onboard SpIRIT.
- Y. Evangelista et al., 2024, *SPIE Conf. Ser.*, 130931Z — The HERMES (High Energy Rapid Modular Ensemble of Satellites) Pathfinder mission.
- G. Baroni et al., 2026, *SPIE Conf. Ser.*, 14146 — In-flight performance and observations of the high-energy HERMES payload onboard SpIRIT and HERMES-PF.

First-author articles (Under Revision)

- **Leone W.** et al., 2026, *A&A* (Under Revision) — Constraints on Lorentz invariance violation from energy-resolved timing of gamma-ray bursts. A technique independent of the Gamma Ray Burst emission mode.